

Vulnerability Assessment Report for a Live Website

Cybersecurity Task 1 – Future Interns

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Internship Name: CyberSecurity Internship

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Assessment Type: Passive Vulnerability
Assessment

Tools Used: Nmap, OWASP ZAP, Browser
DevTools

Introduction :

A Vulnerability Assessment is the process of identifying, analyzing, and reporting security weaknesses in a website or web application. The purpose of this assessment is to evaluate the security posture of the target website using penetration testing and vulnerability scanning tools such as Nmap, OWASP ZAP, and Burp Suite.

Objectives:

- Identify security vulnerabilities in the website
- Analyze risks associated with discovered vulnerabilities
- Recommend mitigation techniques
- Improve the security posture of the web application

Tools Used in the Assessment:

The following tools were used to scan the target website, identify vulnerabilities, and analyze security weaknesses.

Assessment Methodology:

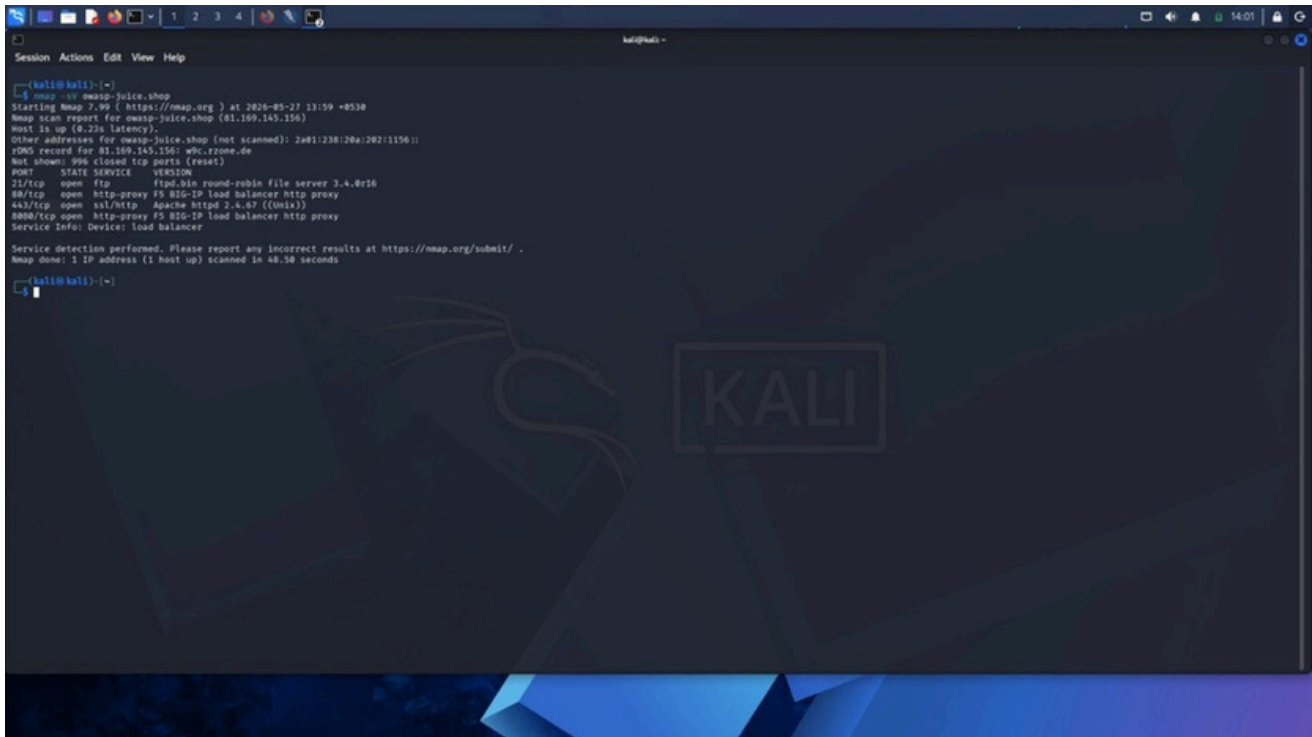
Steps Followed:

1. Reconnaissance and information gathering
2. Port scanning and service detection
3. Web application vulnerability scanning
4. Analysis of discovered vulnerabilities
5. Risk assessment
6. Recommendation and mitigation

The assessment followed a systematic approach to identify vulnerabilities without causing damage to the target system.

Tool Name	Purpose
Kali Linux	Penetration testing operating system
Nmap	Network scanning and port discovery
OWASP ZAP	Web vulnerability scanning
Burp Suite	Intercepting and analyzing web traffic

Kali Linux Setup:



This screenshot shows the Kali Linux environment used during the vulnerability assessment process. Kali Linux contains various penetration testing tools required for ethical hacking and security testing.

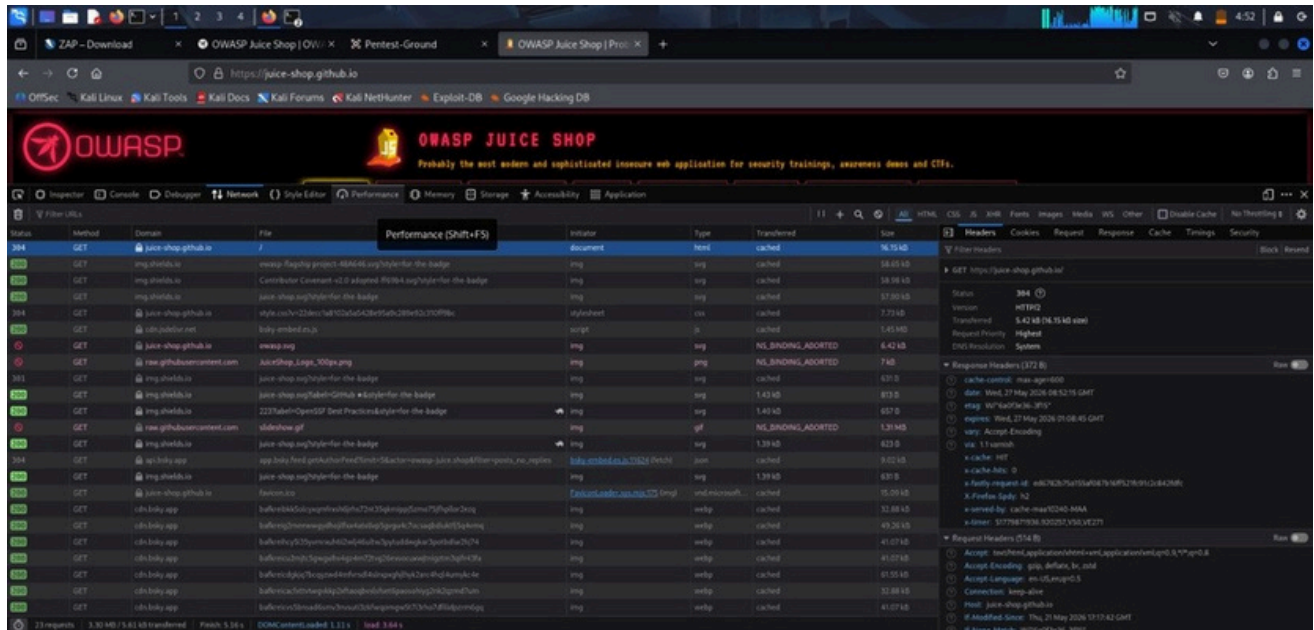
Observed Vulnerabilities:

- Missing Security Headers
- Missing Content Security Policy (CSP)
- Missing Anti-Clickjacking Header
- HTTP Only Site
- X-Content-Type-Options Header Missing

Recommendations:

- Configure secure HTTP headers properly
- Use HTTPS communication
- Implement Content Security Policy (CSP)
- Validate and sanitize user inputs
- Perform regular vulnerability assessments

Burp Suite Vulnerability Scan:



This screenshot displays the Burp Suite vulnerability scanning dashboard. The tool identifies vulnerabilities in the web application and categorizes them based on severity levels such as High, Medium, and Low.

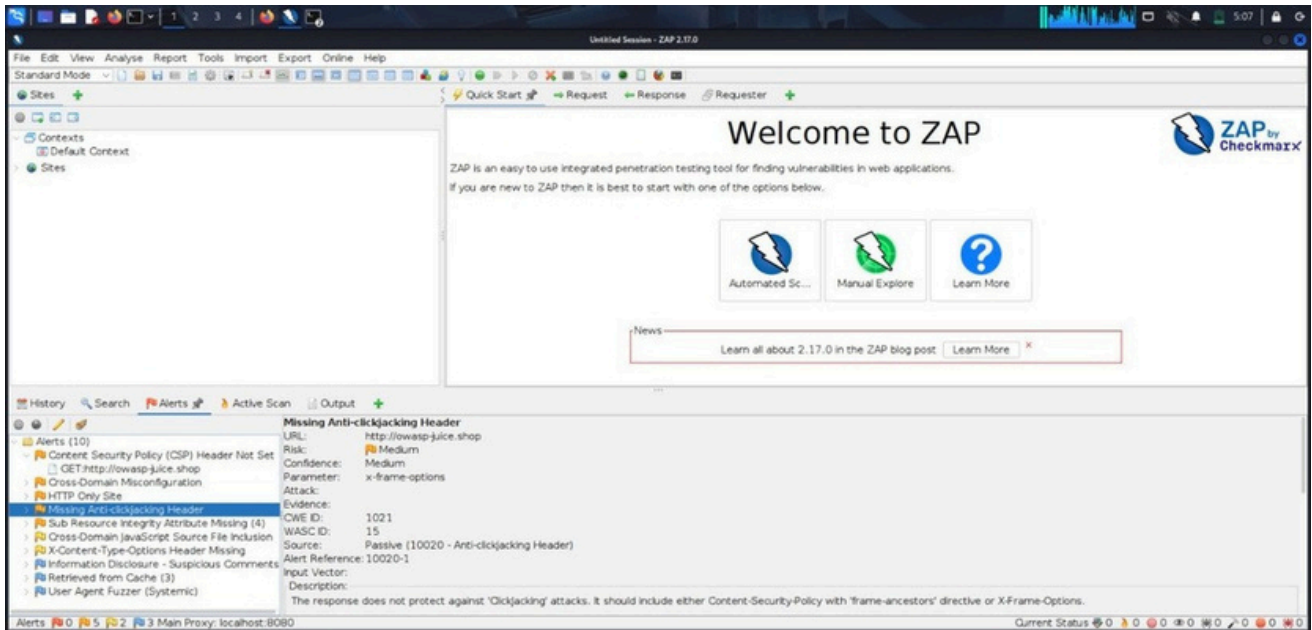
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OWASP ZAP Tool:



This screenshot shows the OWASP ZAP interface used for automated web application vulnerability scanning. The tool helps identify common security vulnerabilities in websites.

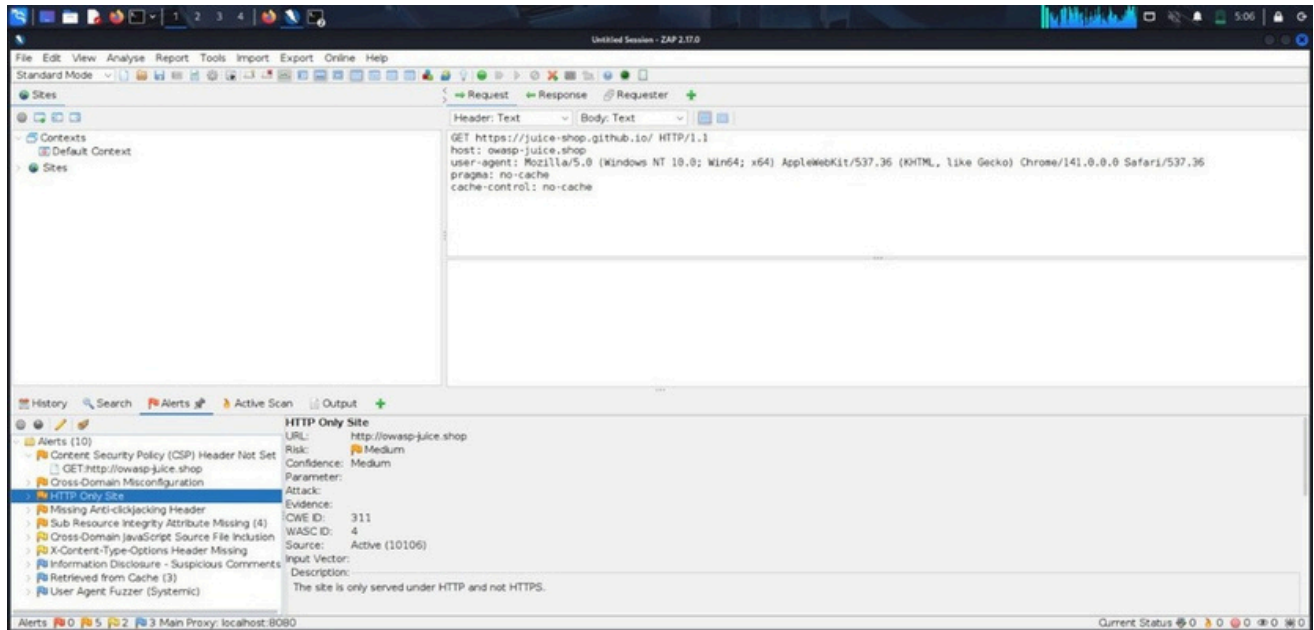
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Vulnerability Alert Analysis:



This screenshot displays detailed vulnerability alerts generated by OWASP ZAP during the assessment process.

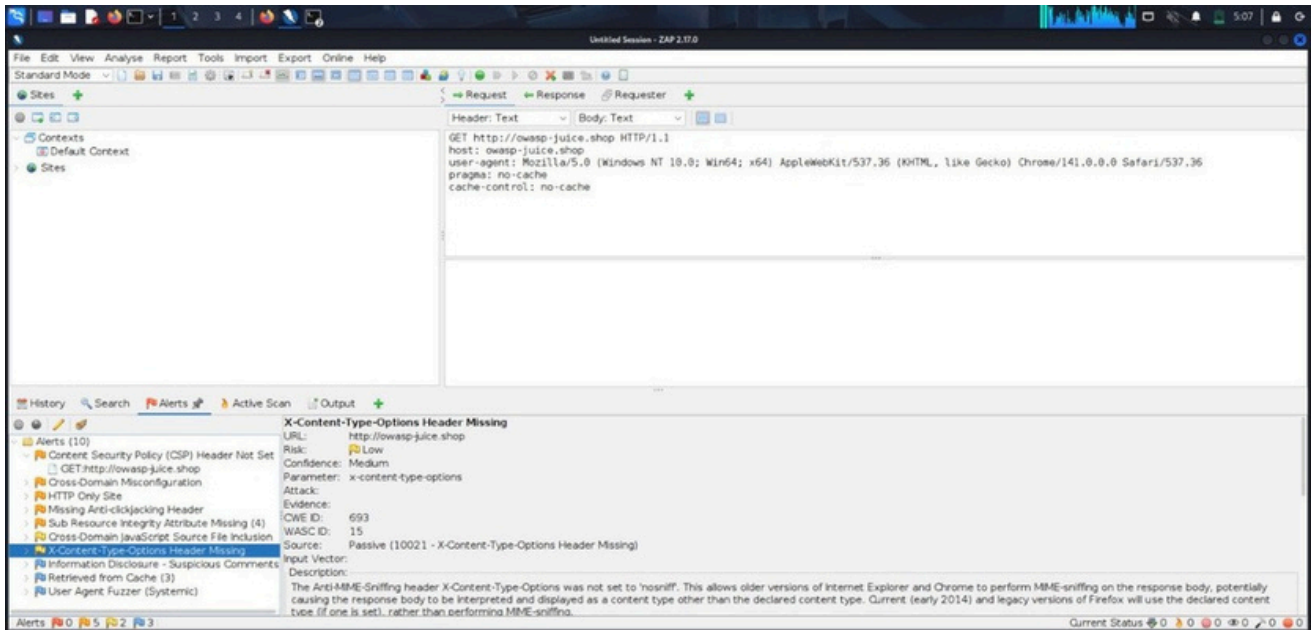
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Security Alert Information:



This screenshot contains additional security alerts and vulnerability details discovered during the scan.

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Risk Assessment:

Vulnerability	Severity	Impact
Missing Security Headers	Medium	Information Disclosure
Insecure Cookies Cross-	High	Session Hijacking Client-
Site Scripting Information	High	side Attacks
Leakage	Medium	Reconnaissance Support
Authentication	High	Unauthorized Access

Security Recommendations:

- Implement HTTPS across the website
- Configure secure HTTP headers
- Use strong authentication and session management
- Regularly update and patch software
- Perform regular vulnerability assessments
- Validate and sanitize user inputs
- Monitor logs and suspicious activities

Conclusion:

The vulnerability assessment successfully identified several security weaknesses in the target web application. Using tools such as Kali Linux, Burp Suite, and OWASP ZAP helped analyze potential threats and risks. Proper remediation and security practices are necessary to protect the application from future attacks.